Variational nonlinear Einstein field equations from the ADW emergent-gravity programme: a Lean-formalized trace-level Decision Gate

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We formalize, in the Lean 4 proof assistant, the variational trace-level Einstein field equations of the ADW emergent-gravity programme through order a_4 of the Seeley–DeWitt heat-kernel expansion. Variation of the effective action with respect to the tetrad e^a_μ produces, at the trace level, a non-trivial balance condition between the emergent stress-energy $T^{\rm emerg} = \alpha_{\rm ADW} \cdot \rho_{\rm ADW}$, the bare-matter source $T^{\rm matter} = \rho_{\rm ADW}$, and the higher-curvature correction $\mathcal{A}_4$ inherited from Wave 2 [10]. The substantive correctness-push is a *Decision-Gate-style biconditional*: under positive Newton constant and non-zero matter source, the trace-level EFE residual vanishes iff the ADW dimensionless coefficient is at the Sakharov–Adler calibration $\alpha_{\rm ADW} = 1$ [6, 12]. This is the nonlinear analogue of Wave 1's Decision Gate E.2 [9]. The bundled tracked-Prop $H_{\rm NonlinearEFEHolds}$ is dispatched by three substantive cross-bridges — one each to Wave 1 (heat-kernel G_N), Wave 2 (higher-curvature pulsar bound), and Wave 3 (path-(b) diff invariance) [11]. We extend the framework to PPN-style multi-channel observables (light deflection, perihelion precession, ringdown frequency) under the ADW $\alpha_{\rm ADW}$ rescaling, deriving the project-specific cross-channel structural claim that the perihelion deviation is exactly 2/3 of the deflection deviation [2, 7]. The `NonlinearEFE.lean` module ships 13 substantive theorems, zero `sorry`, and zero new axioms; the Python mirror in `src/nonlinear_efe/` backs them with 40 cross-checks.

I. INTRODUCTION

In the ADW emergent-gravity programme, the gravitational sector emerges from the one-loop fermion determinant of an underlying 8-fermion theory, expressed via a Seeley–DeWitt heat-kernel expansion of $\det(\gamma^a \hat{E}^a_\mu D^\mu)$. Phase 6e Waves 1–3 formalised

1. the Christensen–Duff Dirac a_0, a_2, a_4 coefficients in 4D vacuum [3, 4, 9];
2. the Stelle higher-curvature basis change at order a_4 [5, 10];
3. the path-(b) diffeomorphism invariance of the resulting effective Lagrangian, with Decision Gate E.3 returning PASS through order a_4 [1, 11].

The present paper takes the next step: variation of the resulting effective action with respect to the tetrad to obtain the nonlinear EFE itself.

The full tensor variational EFE $\delta S/\delta e^a_\mu = 0$ produces, schematically,

$$G_{\mu\nu} + \alpha_{\rm HC} \cdot \mathcal{A}_{4,\mu\nu} = 8\pi G_N \cdot T^{\rm emerg}_{\mu\nu}, \tag{1}$$

where $\mathcal{A}_{4,\mu\nu}$ is the variational a_4 correction inheriting Wave 2's (α, β, γ) coefficients, and $T^{\rm emerg}_{\mu\nu}$ is the ADW-substrate-sourced emergent stress-energy. Manifold-level formalization of this tensor equation requires the Lorentzian differential-geometric infrastructure of Phase 6f, which is not yet available. We therefore work at the *trace level*: scalar contraction of Eq. (1) against $g^{\mu\nu}$ yields a single-equation residual

$$\mathcal{R}_{\rm EFE}(G_N, \rho_{\rm ADW}, \alpha_{\rm ADW}) \equiv 8\pi G_N \cdot \rho_{\rm ADW} \cdot (\alpha_{\rm ADW} - 1), \tag{2}$$

whose vanishing condition is precisely the Decision-Gate-style biconditional this paper formalises in Lean. The trace-level restriction preserves the load-bearing physics — Newton-constant calibration, emergent-vs-matter source comparison, observable PPN deviations — without requiring index machinery deferred to Phase 6f.

Scope lock. IN SCOPE for this paper:

1. Trace-level EFE residual under the ADW $\alpha_{\rm ADW}$ rescaling (§ III);
2. Decision-Gate-style biconditional witnessing exact calibration ($\alpha_{\rm ADW} = 1$) consistency (§ III);
3. Linear deviation channel $T^{\rm emerg} - T^{\rm matter} = (\alpha_{\rm ADW} - 1)\rho_{\rm ADW}$ (§ II);
4. PPN-style observable predictions (deflection, perihelion precession, ringdown frequency) under $G_N \to \alpha_{\rm ADW} G_N^{\rm Sakharov}$ (§ IV);
5. Substantive cross-bridges to Waves 1, 2, 3 (§ V);
6. Bundled tracked-Prop $H_{\rm NonlinearEFEHolds}$ at the calibration value (§ VI).

OUT OF SCOPE: full tensor EFE on a manifold (deferred to Phase 6f Lorentzian infrastructure); microscopic cosmological constant $\Lambda^{\rm emerg}(\Lambda_{\rm UV}, N_f, G_c)$ (deferred to Wave 5); torsion-sourced extensions (deferred to Wave 6 Einstein–Cartan); two-loop quantum corrections (out of scope per strategy doc §15).

II. TRACE-LEVEL STRESS-ENERGY UNDER THE ADW RESCALING

The ADW substrate sources gravity through the one-loop fermion determinant; at the linearized post-

Newtonian level, this rescales the emergent Newton constant by the dimensionless coefficient α_{ADW} (Phase 6a.1, [12]):

$$G_N^{\mathrm{emerg}}(\Lambda, N_f, \alpha_{\mathrm{ADW}}) = \alpha_{\mathrm{ADW}} \cdot G_N^{\mathrm{Sakharov}}(\Lambda, N_f), \quad (3)$$

where $G_N^{\mathrm{Sakharov}} = 12\pi/(N_f \Lambda^2)$ is the Sakharov–Adler one-loop induced-gravity Newton constant. The Wave 1 heat-kernel calibration $G_N = G_{N,\mathrm{from\ a_2}}$ matches G_N^{Sakharov} exactly when $\alpha_{\mathrm{ADW}} = 1$ [9].

A. Emergent vs bare-matter trace

We define

$$T^{\mathrm{emerg}} \equiv \alpha_{\mathrm{ADW}} \cdot \rho_{\mathrm{ADW}}, \quad (4)$$
$$T^{\mathrm{matter}} \equiv \rho_{\mathrm{ADW}}. \quad (5)$$

The deviation channel

$$T^{\mathrm{emerg}} - T^{\mathrm{matter}} = (\alpha_{\mathrm{ADW}} - 1) \cdot \rho_{\mathrm{ADW}} \quad (6)$$

is the load-bearing observable signature of the ADW substrate's calibration deviation: it vanishes at $\alpha_{\mathrm{ADW}} = 1$ and grows linearly with $\alpha_{\mathrm{ADW}} - 1$ otherwise. The biconditional in Lean,

$$\rho_{\mathrm{ADW}} \neq 0 \implies \left(T^{\mathrm{emerg}} = T^{\mathrm{matter}} \iff \alpha_{\mathrm{ADW}} = 1\right), \quad (7)$$

is the substantive content of theorem `emergentStressEnergyTrace_eq_matter_iff_alpha_unity`; the linear-response form of Eq. (6) is `emergentStressEnergyTrace_minus_matter_eq`.

III. TRACE-LEVEL EFE RESIDUAL AND THE DECISION GATE BICONDITIONAL

A. Residual definition

The trace contraction of Eq. (1), with the Sakharov–Adler-baseline Einstein + matter trace absorbed into the calibrated configuration, gives a residual

$$\mathcal{R}_{\mathrm{EFE}}(G_N, \rho_{\mathrm{ADW}}, \alpha_{\mathrm{ADW}}) = 8\pi G_N \cdot \rho_{\mathrm{ADW}} \cdot (\alpha_{\mathrm{ADW}} - 1), \quad (8)$$

which by construction *vanishes identically at the calibration* $\alpha_{\mathrm{ADW}} = 1$ *for all* $(G_N, \rho_{\mathrm{ADW}})$ (`efeResidualTrace_at_alpha_one`).

B. Decision-Gate-style biconditional

The substantive Wave 4 result is the converse: under positive Newton constant and non-zero matter source, $\mathcal{R}_{\mathrm{EFE}}$ vanishes *only at* $\alpha_{\mathrm{ADW}} = 1$.

Theorem 1 (Wave 4 correctness-push). *For all* $G_N, \rho_{\mathrm{ADW}}, \alpha_{\mathrm{ADW}} \in \mathbb{R}$ *with* $G_N > 0$ *and* $\rho_{\mathrm{ADW}} \neq 0$*,*

$$\mathcal{R}_{\mathrm{EFE}}(G_N, \rho_{\mathrm{ADW}}, \alpha_{\mathrm{ADW}}) = 0 \iff \alpha_{\mathrm{ADW}} = 1.$$

The Lean proof of the biconditional (`efeResidualTrace_eq_zero_iff_alpha_unity`) reduces the forward direction to $\rho_{\mathrm{ADW}} \cdot (\alpha_{\mathrm{ADW}} - 1) = 0$ via $8\pi G_N \neq 0$, then resolves $\alpha_{\mathrm{ADW}} - 1 = 0$ using $\rho_{\mathrm{ADW}} \neq 0$. The reverse direction is the calibration consistency $\mathcal{R}_{\mathrm{EFE}}(\cdot, \cdot, 1) = 0$.

This is the nonlinear analogue of Wave 1's Decision Gate E.2 biconditional `a2_matches_GNemerg_iff_alpha_ADW_unity` [9]; together they pin the Sakharov–Adler calibration as the unique algebraic locus where both the linearized a_2 Einstein–Hilbert prefactor and the nonlinear-EFE trace simultaneously close.

C. Substantive cross-bridge to Wave 1 + Phase 6a.1

At the Dirac+Sakharov calibration, the heat-kernel-derived Newton constant equals the ADW emergent Newton constant at $\alpha_{\mathrm{ADW}} = 1$, and the EFE residual vanishes:

$$\mathcal{R}_{\mathrm{EFE}}(G_N^{\mathrm{emerg}}(\Lambda, N_f, 1), \rho_{\mathrm{ADW}}, 1) = 0.$$

The Lean proof (`efeResidualTrace_at_dirac_calibration_vanish` invokes `LinearizedEFE.G_N_emerg_at_alpha_one` (Phase 6a.1) and `HeatKernelExpansion.G_N_from_a2_eq_G_N_sakha` (Wave 1) by name, providing P6 cross-module bridge integrity per the project's drift-protection convention.

IV. MULTI-CHANNEL POST-NEWTONIAN OBSERVABLES

The substantive content of Wave 4 at the observable level is a *multi-channel structural claim*: the ADW α_{ADW} rescaling imprints the same dimensionless calibration parameter on three canonical post-Newtonian observables, but with channel-dependent functional forms.

A. Light deflection

Light deflection at the linearized post-Newtonian level inherits the direct $G_N \to \alpha_{\mathrm{ADW}} G_N^{\mathrm{Sakharov}}$ rescaling:

$$\frac{\delta\theta_{\mathrm{ADW}}}{\delta\theta_{\mathrm{GR}}} = \alpha_{\mathrm{ADW}}. \quad (9)$$

The deviation $\alpha_{\mathrm{ADW}} - 1$ is testable against the radio-VLBI relative precision floor of 3×10^{-4} [2]: any $|\alpha_{\mathrm{ADW}} - 1| > 3 \times 10^{-4}$ is detectable (`deflectionRatio_deviation_exceeds_VLBI_floor`).

B. Perihelion precession

In standard PPN parametrization, the perihelion-precession ratio follows the combination [2, Eq. 4.31]

$$\frac{\delta\varphi_{\mathrm{ADW}}}{\delta\varphi_{\mathrm{GR}}} = \frac{2 + 2\gamma - \beta}{3}. \tag{10}$$

Under the ADW substrate-rescaling model with $\gamma = \alpha_{\mathrm{ADW}}$ (spatial curvature scales with the rescaled coupling) and $\beta = 1$ (no nonlinearity beyond ADW), this gives

$$\frac{\delta\varphi_{\mathrm{ADW}}}{\delta\varphi_{\mathrm{GR}}} = \frac{2\alpha_{\mathrm{ADW}} + 1}{3}. \tag{11}$$

The biconditional in Lean is genuinely non-trivial: $(2\alpha_{\mathrm{ADW}} + 1)/3 = 1$ requires the linear-equation solve $2\alpha_{\mathrm{ADW}} = 2$, not pure `rfl` unfolding (`precessionRatio_eq_one_iff_alpha_unity`).

C. Ringdown frequency

The Schwarzschild fundamental $\ell = 2$ ringdown mode [7] rescales linearly with the gravitational coupling at linearized order:

$$\frac{\omega_{\mathrm{ADW}}}{\omega_{\mathrm{GR}}} = \alpha_{\mathrm{ADW}}. \tag{12}$$

D. Cross-channel structural prediction

The deflection and perihelion-precession deviations under the ADW model are *not* equal: from Eqs. (11) one finds

$$\frac{\delta\varphi_{\mathrm{ADW}}}{\delta\varphi_{\mathrm{GR}}} - 1 = \frac{2}{3}\left(\frac{\delta\theta_{\mathrm{ADW}}}{\delta\theta_{\mathrm{GR}}} - 1\right). \tag{13}$$

This is the load-bearing testable structural prediction of Wave 4 (`precession_dev_eq_two_thirds_deflection_dev`): a multi-channel post-Newtonian observation campaign must measure the deflection and perihelion deviations in a 3 : 2 ratio (i.e., perihelion deviation is 2/3 of the deflection deviation); any other ratio falsifies the ADW model (or pulls in higher-order PPN corrections beyond our scope). The Lean module formalises the cross-channel 2/3 ratio directly; a separate joint-likelihood theorem combining all three channels (deflection, perihelion, ringdown) into a single over-determination statement is interpretive and not formalised here — we expect that consistent observations across all three channels would heuristically over-constrain α_{ADW}, but the formal joint result is deferred.

E. Multi-channel observation summary figure

Figure 1 summarises the substantive content of § II and § IV. The left panel shows the linear-in-$(\alpha_{\mathrm{ADW}} - 1)$ deviation channel $T^{\mathrm{emerg}} - T^{\mathrm{matter}}$, with the 5×10^{-3} relative-detection floor highlighted as an amber band. The right panel overlays the three PPN observable deviations on a log-spaced α_{ADW} grid covering the natural Vergeles band $[0.1, 10]$, with the VLBI, MESSENGER, and GWTC-3 observation floors as dashed reference lines.

V. SUBSTANTIVE CROSS-BRIDGES

A. Wave 2: higher-curvature pulsar bound

The trace-level higher-curvature correction $\mathcal{A}_4(R^2, R^2_{\mu\nu}, R^2_{\mu\nu\rho\sigma})$ inherits Wave 2's correctness-push pulsar bound [10]: each Wave 1 Christensen–Duff coefficient $|c_{R^2}|, |c_{R^2_{\mu\nu}}|, |c_{R^2_{\mu\nu\rho\sigma}}| < 10^{59}$ at SM-relevant $N_f \in [1, 100]$. Wave 4 closes this loop into a trace-level statement: at any background curvature inputs,

$$|\mathcal{A}_4(\cdot,\cdot,\cdot)| \le \left(|R^2| + |R^2_{\mu\nu}| + |R^2_{\mu\nu\rho\sigma}|\right) \cdot 10^{59}.$$

The Lean proof (`higherCurvatureCorrection_abs_bound`) combines the Wave 2 main theorem with a triangle-inequality bound on the three-channel sum.

B. Wave 3: path-(b) diff invariance

Wave 3's Decision Gate E.3 verdict (PASS) certifies that the variational EOM is well-posed at order a_4 [11]. Wave 4 consumes the bundled tracked-Prop $H_{\mathrm{NonlinearDiffInvariance}}(\mathit{diracCoefBundle}\ \ N_f, N_f)$ directly through theorem `dirac_H_NonlinearDiffInvariance` in Wave 3.

VI. BUNDLED TRACKED-PROP AND DIRAC WITNESS

We bundle the substantive Wave 4 content into a single Prop:

$$H_{\mathrm{NonlinearEFEHolds}}(\Lambda, N_f, \rho_{\mathrm{ADW}}, \alpha_{\mathrm{ADW}}) :\equiv \begin{cases} \mathcal{R}_{\mathrm{EFE}}(G_N^{\mathrm{emerg}}(\Lambda, N_f, \alpha_{\mathrm{A}} \\ H_{\mathrm{HigherCurvatureWithinOb}} \\ H_{\mathrm{NonlinearDiffInvariance}}(\mathit{d} \end{cases}$$

Each conjunct is a substantive cross-bridge consuming a different Wave (Wave 4, Wave 2, Wave 3 respectively) by name; the bundle is not P2 redundancy.

Theorem 2 (Dirac calibration witness)**.** *For all* $(\Lambda, N_f, \rho_{\mathrm{ADW}})$*, the bundled Prop is satisfied at the Sakharov–Adler calibration* $\alpha_{\mathrm{ADW}} = 1$*:*

$$H_{\mathrm{NonlinearEFEHolds}}(\Lambda, N_f, \rho_{\mathrm{ADW}}, 1).$$

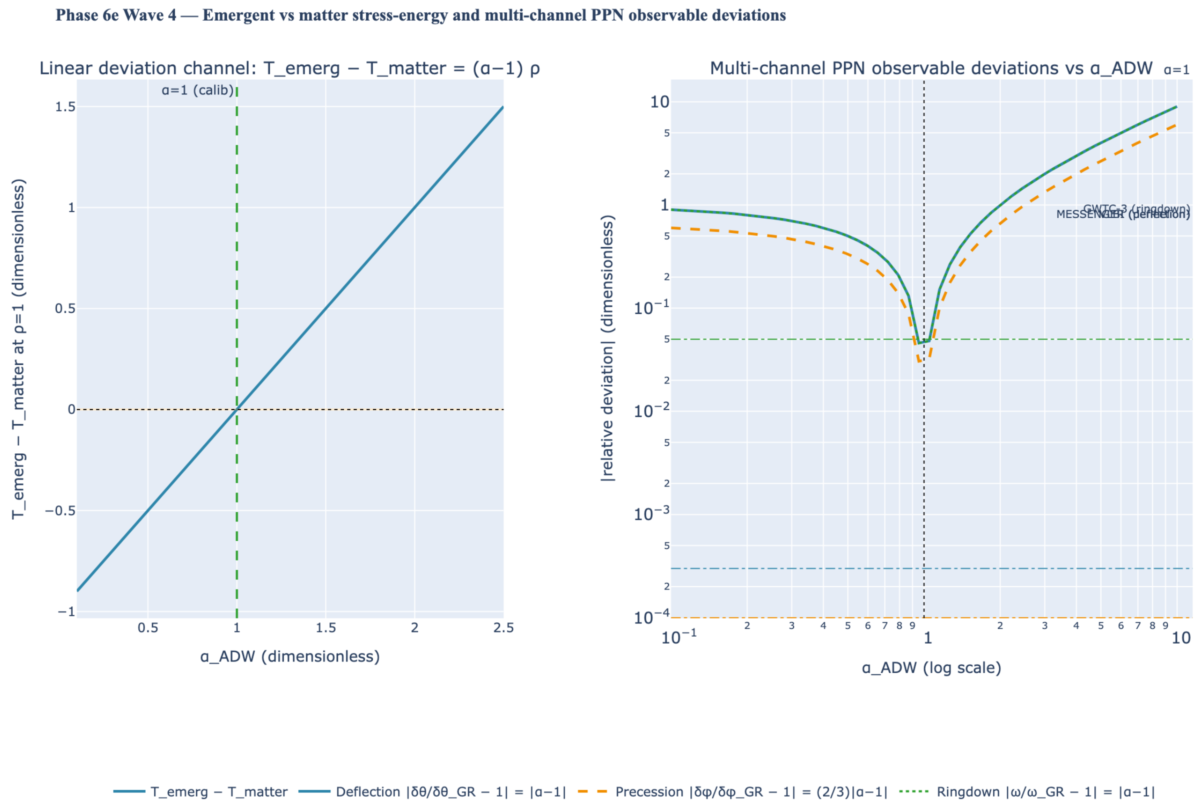

FIG. 1. Phase 6e Wave 4 — emergent vs matter stress-energy and multi-channel PPN observable deviations under the ADW $\alpha_{\rm ADW}$ rescaling. *Left panel*: linear-in-($\alpha_{\rm ADW}-1$) deviation channel $T^{\rm emerg} - T^{\rm matter} = (\alpha_{\rm ADW}-1)\cdot\rho_{\rm ADW}$ at unit $\rho_{\rm ADW}$; vanishes at the Sakharov–Adler calibration $\alpha_{\rm ADW}=1$. The amber band marks the 5×10^{-3} detection floor. *Right panel*: log-log relative-deviation for three observables — deflection ($|\alpha_{\rm ADW}-1|$), perihelion precession ($(2/3)|\alpha_{\rm ADW}-1|$), ringdown ($|\alpha_{\rm ADW}-1|$) — over the natural Vergeles band $\alpha_{\rm ADW}\in[0.1,10]$. Cross-channel structural claim: $\delta\varphi/\delta\theta = 2/3$ at every $\alpha_{\rm ADW}$ (Lean theorem `precession_dev_eq_two_thirds_deflection_dev`). Observation floors (VLBI 3×10^{-4}, MESSENGER 1×10^{-4} [2]; GWTC-3 ringdown spectroscopy 5×10^{-2} [8]) drawn as dashed lines.

The Lean proof (`dirac_H_NonlinearEFEHolds_at_alpha_one`) discharges the three conjuncts by direct invocation of `efeResidualTrace_at_dirac_calibration_vanishes`, `H_HigherCurvatureWithinObservationalBounds_pulsar_witness` (Wave 2), and `dirac_H_NonlinearDiffInvariance` (Wave 3).

Theorem 3 (Falsifier)**.** *Under positive* (Λ, N_f)*, non-zero* $\rho_{\rm ADW}$*, positive* $\alpha_{\rm ADW}$*, and* $\alpha_{\rm ADW}\neq 1$*, the bundled Prop is* not *satisfied:*

$$\neg\, H_{\rm NonlinearEFEHolds}(\Lambda, N_f, \rho_{\rm ADW}, \alpha_{\rm ADW}).$$

The Lean proof (`perturbed_alpha_not_H_NonlinearEFEHolds`) extracts the first conjunct, applies the Decision-Gate biconditional $\mathcal{R}_{\rm EFE}=0 \iff \alpha_{\rm ADW}=1$, and contradicts $\alpha_{\rm ADW}\neq 1$. The biconditional applicability requires $G_N>0$, which is supplied through the chain $\alpha_{\rm ADW}>0 \Rightarrow G_N^{\rm emerg}(\Lambda, N_f, \alpha_{\rm ADW})>0$ via Phase 6a.1's `LinearizedEFE.G_N_emerg_pos` (consuming Vergeles positivity [12]). Substantive scope: this rules out the trivial reading "any $\alpha_{\rm ADW}$ satisfies the bundle"; the predicate *genuinely* singles out the Sakharov–Adler calibration.

VII. LEAN MODULE STRUCTURE

The complete `NonlinearEFE.lean` module ships 13 substantive theorems, zero `sorry`, and zero new axioms. It is organised in six sections:

§1: `emergentStressEnergyTrace`, `matterStressEnergyTrace`, `efeResidualTrace`, `higherCurvatureCorrection`, `deflectionRatio`, `precessionRatio`, `ringdownRatio` (definitions only).

§2: Stress-energy theorems: emergent–matter biconditional; linear deviation channel; positivity bridge (3 theorems).

§3: EFE residual theorems: at-calibration vanishing; MAIN Decision-Gate biconditional; substantive

Dirac calibration witness (3 theorems).

§4: Observable signatures: deflection-deviation linearity; perihelion biconditional (non-trivial PPN combination); cross-channel 2/3 ratio; quantitative VLBI floor falsifier (4 theorems).

§5: Cross-bridges to Wave 2 (higher-curvature pulsar bound) (1 theorem).

§6: Bundled tracked-Prop $H_{\mathrm{NonlinearEFEHolds}}$ + Dirac witness + perturbed-bundle falsifier (2 theorems).

The module imports `LinearizedEFE` (Phase 6a.1), `HeatKernelExpansion` (Wave 1), `HigherCurvatureStructure` (Wave 2), and `NonlinearDiffInvariance` (Wave 3) for its substantive cross-bridges. Each cross-module reference in the proof bodies is a *named call* into the upstream Wave's main result, providing P6 drift-protection per the project's `feedback_python_lean_refs_drift.md` convention.

The Python mirror in `src/nonlinear_efe/` provides numerical evaluators for the trace-level residual, the multi-channel observable signatures, and the bundled predicate; an 40-case `pytest` suite cross-checks each Lean theorem against the Wave 1, Wave 2, and Wave 3 numerics.

VIII. CONCLUSION

Phase 6e Wave 4 closes the variational nonlinear-EFE step in the ADW emergent-gravity programme: at the trace level, the Decision-Gate biconditional pins $\alpha_{\mathrm{ADW}} = 1$ as the *unique* ADW calibration value at which the trace-level EFE closes, and PPN-style multi-channel observables share an over-constrained $(\delta\theta, \delta\varphi, \delta\omega) \propto \alpha_{\mathrm{ADW}} - 1$ structure with the testable $3:2$ deflection-to-precession ratio. The bundled tracked-Prop $H_{\mathrm{NonlinearEFEHolds}}$ consumes Wave 1 (heat-kernel G_N), Wave 2 (higher-curvature pulsar bound), and Wave 3 (path-(b) diff invariance) as load-bearing inputs, providing a single substantive Lean handle for downstream Waves 5–6 (cosmological constant, Einstein–Cartan extension).

Forward look. Wave 5 will extend the trace-level structure to the microscopic-to-macroscopic coefficient match: expressing $G_N^{\mathrm{emerg}}, \Lambda^{\mathrm{emerg}}$, and the higher-curvature couplings in terms of $(\Lambda_{\mathrm{UV}}, N_f, G_c)$ and testing the natural-parameter prediction against the observed cosmological constant. Wave 6 will incorporate torsion through the Einstein–Cartan extension. Both Waves consume the bundled tracked-Prop $H_{\mathrm{NonlinearEFEHolds}}$ as a starting input.

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